

SENTINEL D2

Zero-Trust Threat Intelligence & Active Containment

Team: RootCause

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THE BATTLEFIELD REALITY

Problem Statement: Alert Fatigue & Delayed Response

Defence analysts are currently overwhelmed by the sheer volume of daily security alerts generated by SIEM systems, satellite feeds, and cyber sensors. The noise makes it impossible for a human team to manually read, correlate, and investigate every single event.

1. The "Needle in a Haystack"

A genuine, catastrophic Advanced Persistent Threat (APT) can easily be missed among the thousands of routine network pings and false positives.

2. Resource Drain

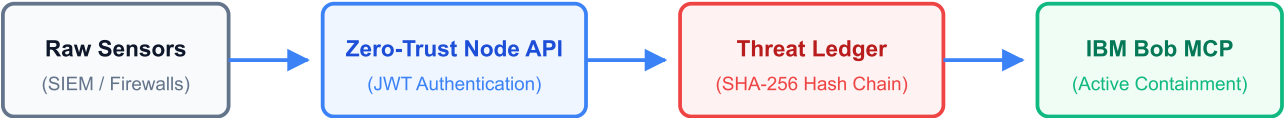
Chasing false alarms wastes critical analyst time. By the time human operators correlate the data, adversaries have already pivoted through the network.

The Commander's Dilemma: Military commanders cannot make tactical decisions by reading raw server logs. They require structured, *Bottom Line Up Front (BLUF)* summaries and immediate containment options to secure the perimeter.

THE SENTINEL D2 SOLUTION

System Architecture & Active Defense

Sentinel D2 acts as an intelligent, automated filter between raw security feeds and military commanders. It utilizes a multi-phase correlation engine to group isolated events into unified attack campaigns.



Zero-Trust Ingestion

Instead of blindly accepting logs, our Node.js API requires JWT authentication. This prevents adversaries from flooding the dashboard with fake alerts as a distraction.

Immutable Ledger

Critical alerts are written to a cryptographic ledger using SHA-256 chaining. If an attacker alters a log to cover their tracks, the hash breaks, exposing the tampering.

STRATEGIC ADVANTAGE

Key Features & Future Scope

Capability	Traditional SIEM Dashboards	Sentinel D2 Advantage
Data Ingestion	Accepts all incoming plaintext logs.	Zero-Trust JWT API rejects unauthorized payloads.
Log Integrity	Vulnerable to hacker deletion/alteration.	SHA-256 cryptographic hash chaining locks records.
Incident Response	Passive text summaries only.	IBM Bob MCP executes automated containment scripts.
Threat Context	Isolated IP addresses and pings.	Automated mapping to MITRE ATT&CK framework.

Technical Limitations & Next Steps:

For Phase 1 (Online Submission), the backend operates on simulated static data to guarantee 100% reproducibility for the evaluation team. If selected for the offline round at CHARUSAT, the architecture is designed to seamlessly integrate live WebSockets and a persistent PostgreSQL database for real-time threat streaming.